BioHackrXiv

# BioHackEU23 report: Template for the very long title

**First Author**[1] **and Last Author** [2]

**1** First Affiliation **2** ELIXIR Europe

**BioHackathon series:**
BioHackathon Europe 2023
Barcelona, Spain, 2023
*Project 26*

**Submitted:** 12 Nov 2026

**License:**
Authors retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC-BY).

Published by BioHackrXiv.org

## Introduction

As part of the BioHackathon Europe 2023, we here report...

### Meeting information

If you want to submit a preprint to BioHackrXiv, first check if your meeting is registered. You can find a list of meetings here. If your meeting is missing, please contact your meeting organizers. The above list also provides information on the YAML fields with information about the meeting.

The following fields need to be given:

```
biohackathon_name: "BioHackathon Europe 2023"
biohackathon_url:   "https://biohackathon-europe.org/"
biohackathon_location: "Barcelona, Spain, 2023"
group: Project 26
git_url: https://github.com/yourOrganization/your_report_repo
```

The BioHackrXiv meeting pages provide content to use for the first three fields. The `git_url:` field must have the link to the GitHub repository with your preprint (draft).

### Author information

Information about the authors is given in the YAML format at the top of this template. For authors you provide their names, their affiliations. That is the minimum, but as BioHackrXiv is moving to a situation where more metadata is shared, and used by, for example, EuropePMC, adding additional information ie encouraged.

BioHackathons is about hacking together, and the minimal number of authors for reports is two. This makes a minimal example look like this:

```
authors:
  - name: First Author
    affiliation: 1
  - name: Last Author
    affiliation: 2
affiliations:
  - name: First Affiliation
    index: 1
  - name: ELIXIR Europe
    index: 2
```

#### Author identifiers

Ideally, authors provide their ORCID identifier. For affiliations, It is added with the `orcid:` field. So, and author record would look like this:

```yaml
authors:
  - name: First Author
    affiliation: 1
    orcid: 0000-0000-0000-0000
```

#### Research Organization Registry identifiers

Matching the author identifier, the affiliations can be further specified with the Research Organization Registry (ROR) identifier. For example, this is the affiliation identifier can be added with the `ror:` field:

```yaml
affiliations:
  - name: ELIXIR Europe
    ror: 044rwnt51
    index: 2
```

#### Contributor Role Taxonomy

A last feature since is minimal support for the Contributor Role Taxonomy (CRediT). You can specify the role of authors in writing the report with the `role:` field. However, the authors are responsible for selection the right terms from CRediT. An example looks like this:

```yaml
authors:
  - name: First Author
    affiliation: 1
    orcid: 0000-0000-0000-0000
    role: Conceptualization, Writing - review & editing
```

#### A full examples

A full example then has this structure:

```yaml
authors:
  - name: First Author
    affiliation: 1
    role: Writing - original draft
  - name: Last Author
    orcid: 0000-0000-0000-0000
    affiliation: 2
    role: Conceptualization, Writing - review & editing
affiliations:
  - name: First Affiliation
    index: 1
  - name: ELIXIR Europe
    ror: 044rwnt51
    index: 2
```

## Formatting

This document use Markdown and you can look at this tutorial.

### Subsection level 2

Please keep sections to a maximum of only two levels.

## Tables

Tables can be added in the following way, though alternatives are possible:

```
Table: Note that table caption is automatically numbered and should be
given before the table itself.

| Header 1 | Header 2 |
| -------- | -------- |
| item 1 | item 2 |
| item 3 | item 4 |
```

This gives:

**Table 1:** Note that table caption is automatically numbered and should be given before the table itself.

| Header 1 | Header 2 |
|---|---|
| item 1 | item 2 |
| item 3 | item 4 |

## Figures

A figure is added with:

```
![Caption for BioHackrXiv logo figure](./biohackrxiv.png)
```

This gives:




**Figure 1:** Caption for BioHackrXiv logo figure

Figures can be scaled by adding the width or height to the Markdown like this:

```
![Caption for BioHackrXiv logo figure](./biohackrxiv.png){ width=50px }
```

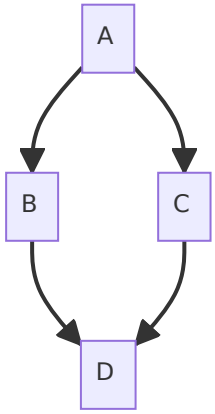
You can add cross references to figures by adding a LaTeX `\label{figureCode}` to the label of the Markdown figure and then use `\ref{figureCode}` to cite it:

```
![Caption for BioHackrXiv logo figure \label{figureCode}](./biohackrxiv.png){ widt
```

This way, we can cite Figure 1.

## Mermaid graphs

Mermaid provides a solution to put diagrams in Markdown. BioHackrXiv added support for this in 2026. This is an example of embedding a graph:




## Other main section on your manuscript level 1

Lists can be added with:

1. Item 1
2. Item 2

## Citation Typing Ontology annotation

You can use CiTO annotations, as explained in this BioHackathon Europe 2021 write up and this CiTO Pilot. Using this template, you can cite an article and indicate *why* you cite that article, for instance DisGeNET-RDF (Queralt-Rosinach et al., 2016).

The syntax in Markdown is as follows: a single intention annotation looks like `[@usesMethodIn:Krewinkel2017]`; two or more intentions are separated with colons, like `[@extends:discusses:Nielsen2017Scholia]`. When you cite two different articles, you use this syntax: `[@citesAsDataSource:Ammar2022ETL; @citesAsDataSource:Arend2022BioHackEU22]`.

Possible CiTO typing annotation include:

- citesAsDataSource: when you point the reader to a source of data which may explain a claim
- usesDataFrom: when you reuse somehow (and elaborate on) the data in the cited entity
- usesMethodIn
- citesAsAuthority
- citesAsEvidence
- citesAsPotentialSolution
- citesAsRecommendedReading
- citesAsRelated
- citesAsSourceDocument
- citesForInformation
- confirms
- documents
- providesDataFor
- obtainsSupportFrom
- discusses
- extends
- agreesWith
- disagreesWith
- updates

There is a general `cites` intention, but this is already implied and should be left out.

# Results

## Proposed Use Cases

## Proposed Use Cases

Bashir: I propose three use cases using NORMAN-SLE, ExpoPortal ontologies, and scientific publications, initially focusing on PFAS.

#### Bashir: Use case 1. Ontology Gap Detection and Improvement

**Question:** Can scientific publications help identify missing concepts and relationships in exposome ontologies?

**Approach:** - Extract chemical, exposure, and health concepts from scientific publications using AI. - Compare extracted information with ontologies in ExpoPortal. - Identify missing concepts, relationships, and cross-ontology mappings.

**Expected Output:** Potential ontology gaps and evidence-based suggestions for improvement.

#### Bashir: Use case 2. Scientific Research Gap Explorer

**Question:** Which chemicals and exposure-related health effects need further scientific research?

**Approach:** - Explore NORMAN-SLE chemicals and related concepts using ExpoPortal ontologies. - Analyze scientific publications to assess available evidence. - Identify topics with limited, missing, or conflicting evidence.

**Expected Output:** Potential research gaps and suggested research questions.

#### Bashir: Use case 3. Chemical Exposure & Health Knowledge Graph

**Question:** How can we connect chemicals, exposure pathways, and health effects using multiple ontologies?

**Approach:** - Link NORMAN-SLE chemicals to relevant ExpoPortal ontologies. - Explore environmental sources, exposure pathways, biological processes, and health outcomes. - Connect relationships to supporting publications.

**Expected Output:** An interactive knowledge graph with traceable scientific evidence.

### Fernando: Use case 4: Impact of Air Pollution on Health Outcomes and Transcriptomic/Proteomic Profiles

- Integrate phenotypic, exposure, demographic, transcriptomic, and proteomic data.
- Map concepts/entities to ExpoPortal ontologies.
- Identify ontology gaps after mapping.
- Apply neuro-symbolic AI to identify associations between the expomosome and BMI/asthma or omic expression levels.

# Discussion

## Bashir: Evaluation and Future Work

- Evaluate ontology coverage and mappings.
- Discuss limitations and missing evidence.

- Explore dataset and publication integration.
- Propose future ExpoPortal improvements. ...


## Acknowledgements

...


# References

## Author contributions

First Author: Writing – original draft; Last Author: Conceptualization, Writing – review & editing.

## References


Queralt-Rosinach, N., Pinero, J., Bravo, À., Sanz, F., & Furlong, L. I. (2016). DisGeNET-RDF: harnessing the innovative power of the Semantic Web to explore the genetic basis of diseases. *Bioinformatics*, *32*(14), 2236–2238. https://doi.org/10.1093/bioinformatics/btw214 **[cito:citesAsAuthority]**.

## Appendices

If you want the Appendix (-ces) to show up after the references, wrap them in after the header, like done in this Markdown file. Look at the source to see the exact structure.